INCH-POUND

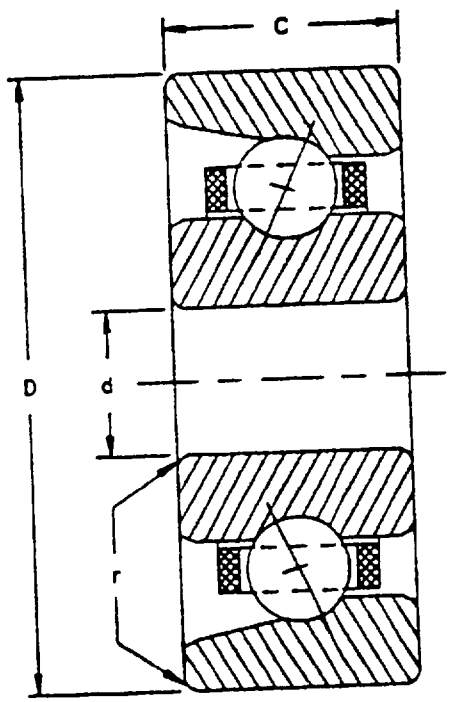
MIL-B-81793/20
15 March 1993

MILITARY SPECIFICATION SHEET

BEARING, BALL, ANNULAR, FOR INSTRUMENTS AND PRECISION ROTATING COMPONENTS, ANGULAR CONTACT, UNFLANGED, NON-SEPARABLE AND COUNTERBORED OUTER RING, CORROSION RESISTANT STEEL, ABEC 7P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-81793



BEARING CONFIGURATIONS

FSC 3110

AMSC N/A

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REQUIREMENTS

1. Design. All bearings described by this specification sheet shall be angular contact instrument bearings with an unflanged, non-separable and counterbored outer ring configuration.

2. Material. The ball and ring material for these bearings shall be corrosion resistant steel 440C (UNS S44004) conforming to QQ-S-763.

3. Tolerance class. The tolerance class shall be in accordance with the ABEC 7P tolerance tables of ANSI/AFBMA Standard 12.2. This tolerance class shall apply to all bearing sizes listed in Table I of this specification sheet.

4. Boundary dimensions. The boundary dimensions shall be as specified by the dash number (DASH NO) in Table I.

5. Retainer. The retainer shall be as specified by the part number designator (PN DES) in Table II.

6. Closures. Closures are not available for this type of bearing. Part number designator N shall be used as shown in Table III.

7. Radial internal clearance or contact angle. Radial internal clearance or contact angle shall be as specified by the part number designator in Table IV.

8. Calibration. Calibration shall be as specified by the part number designator in Table V.

9. Performance Test. The performance test shall be as specified by the part number designator in Table VI.

10. Lubrication.

10.1. Lubricant. The lubricant shall be in accordance with the specification specified by the part number designator in table VII.

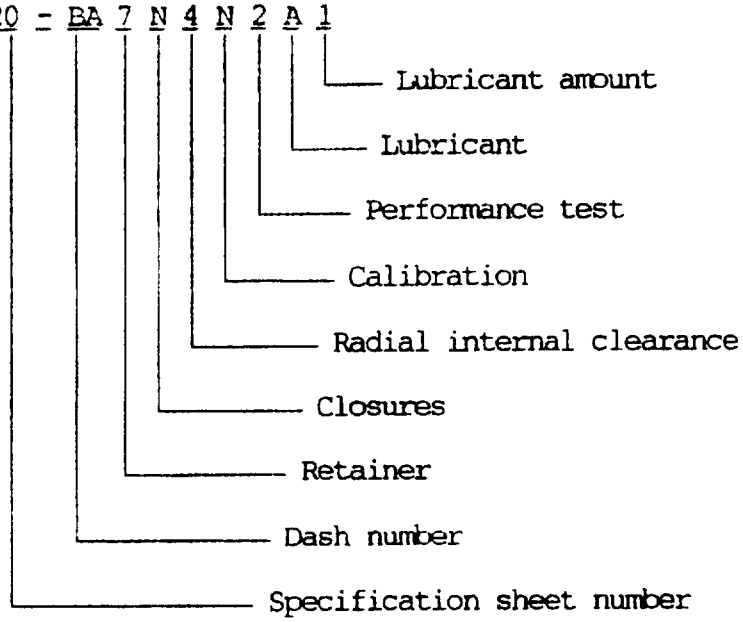
10.2. Lubricant amount. The amount of lubricant shall be as specified by the part number designator in Table VIII.

10.3. Barrier coating. Barrier coating shall be applied to bearings as specified by the part number designator in Table VII.

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11. Part number. The part number consists of the following: prefix letter M; general specification number; number of this specification sheet; dash number; and characters for retainer, closures, radial internal clearance or contact angle, calibration, performance test, lubricant, and lubricant amount.

EXAMPLE: M 81793/ 20 - BA 7 N 4 N 2 A 1



M81793/20-BA7N4N2A1 indicates - Bore 0.0469"; outside diameter 0.1562"; width 0.0937"; radius 0.003"; retainer, phenolic laminate; closures, none; contact angle, greater than 14° but less than 20°; no calibration; performance test, starting torque - Grade A; lubricant, MIL-L-6085; lubricant amount, manufacturer's standard.

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TABLE I. Boundary dimensions (inches).

DASH NO	BORE	OD	WIDTH	RADIUS
	d	D	C	r 1/ 2/
-AA	0.0400	0.1250	0.0469	0.003
-BA	0.0469	0.1562	0.0625	0.003
-BB	0.0469	0.1562	0.0937	0.003
-CA	0.0550	0.1875	0.0781	0.003
-CB	0.0550	0.1875	0.1094	0.003
-DA	0.0781	0.2500	0.0937	0.003
-DB	0.0781	0.2500	0.1406	0.003
-EA	0.0937	0.1875	0.0625	0.003
-EB	0.0937	0.1875	0.0937	0.003
-EC	0.0937	0.2500	0.0625	0.003
-ED	0.0937	0.2500	0.0937	0.003
-EE	0.0937	0.2500	0.1094	0.003
-EF	0.0937	0.2883	0.0625	0.003
-EG	0.0937	0.3125	0.0625	0.005
-EH	0.0937	0.3125	0.1094	0.003
-EJ	0.0937	0.3125	0.1406	0.003
-EK	0.0937	0.4100	0.1094	0.003
-FA	0.1250	0.2500	0.0937	0.003
-FB	0.1250	0.2500	0.1094	0.003
-FC	0.1250	0.3125	0.1094	0.003
-FD	0.1250	0.3125	0.1406	0.003
-FE	0.1250	0.3750	0.1094	0.005
-FF	0.1250	0.3750	0.1406	0.005
-FG	0.1250	0.3750	0.1562	0.012
-FH	0.1250	0.4100	0.0937	0.003
-FJ	0.1250	0.4100	0.1094	0.003
-FK	0.1250	0.4250	0.0937	0.003
-FL	0.1250	0.4250	0.1094	0.003
-FM	0.1250	0.4375	0.1094	0.003
-FN	0.1250	0.4375	0.1406	0.003
-FP	0.1250	0.5000	0.1094	0.003
-FR	0.1250	0.5000	0.1719	0.012
-FT	0.1250	0.7500	0.1250	0.010

1/ Maximum shaft or housing fillet radius that bearings corners will clear.

2/ Narrow face of rings shall have a maximum "r" value of one half the value listed provided this quantity does not fall below 0.003 inches.

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TABLE I. Boundary dimensions (inches) - Continued.

DASH NO	BORE	OD	WIDTH	RADIUS
	d	D	C	r 1/ 2/
-GA	0.1562	0.3125	0.1094	0.003
-GB	0.1562	0.3125	0.1250	0.003
-HA	0.1875	0.3125	0.1094	0.003
-HB	0.1875	0.3125	0.1250	0.003
-HC	0.1875	0.3750	0.1094	0.003
-HD	0.1875	0.3750	0.1250	0.003
-HE	0.1875	0.4100	0.1094	0.003
-HF	0.1875	0.4250	0.1094	0.003
-HG	0.1875	0.4375	0.1094	0.003
-HH	0.1875	0.5000	0.1094	0.003
-HJ	0.1875	0.5000	0.1562	0.012
-HK	0.1875	0.5000	0.1960	0.012
-HL	0.1875	0.7435	0.1960	0.012
-HM	0.1875	0.7500	0.1960	0.012
-HN	0.1875	0.8750	0.1960	0.012
-JA	0.2500	0.3750	0.1250	0.003
-JB	0.2500	0.5000	0.1094	0.003
-JC	0.2500	0.5000	0.1250	0.005
-JD	0.2500	0.5000	0.1875	0.005
-JE	0.2500	0.6250	0.1960	0.012
-JF	0.2500	0.7500	0.1960	0.012
-JG	0.2500	0.7500	0.2188	0.016
-JH	0.2500	0.7500	0.2812	0.016
-JJ	0.2500	0.8750	0.1960	0.012
-JK	0.2500	1.0000	0.1960	0.012
-JL	0.2500	1.0480	0.1960	0.012
-KA	0.3125	0.5000	0.1562	0.005
-KB	0.3125	0.6250	0.1562	0.010
-LA	0.3750	0.8750	0.2188	0.016
-LB	0.3750	0.8750	0.2812	0.016
-MA	0.5000	0.8750	0.2188	0.016
-MB	0.5000	0.8750	0.2812	0.016
-MC	0.5000	1.1250	0.2500	0.016
-MD	0.5000	1.1250	0.3125	0.016
-NA	0.6250	1.3750	0.2812	0.031
-NB	0.6250	1.3750	0.3438	0.031

1/ Maximum shaft or housing fillet radius that bearing corners will clear.

2/ Narrow face of rings shall have a maximum "r" value of one half the value listed provided this quantity does not fall below 0.003 inches.

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TABLE II. Retainer.

PN DES	TYPE
6	PTFE TOROIDS 1/
7	PHENOLIC TYPE LAMINATE 2/, 4/
8	VACUUM IMPREGNATED PHENOLIC 3/, 4/
9	NON-POROUS, NON-METALLIC CROWN 4/

1/ PTFE (Polytetrafluoroethylene).
2/ Phenolic or other porous non-metallic material.
3/ Phenolic or other porous non-metallic material saturated with lubricant. Selection of this choice dictates choosing manufacturer's standard amount of lubricant.
4/ Used for high speed applications.

TABLE III. Closures.

PN DES	NUMBER	TYPE
N	NONE	NONE

TABLE IV. Radial internal clearance or contact angle.

PN DES	RANGES
1	0.0001" to 0.0003"
2	0.0002" to 0.0005"
3	14° OR LESS 1/
4	GREATER THAN 14° BUT LESS THAN 20° 1/
5	0.0005" to 0.0008"
6	GREATER THAN 20° 1/
8	0.0008" to 0.0011"

1/ Angular contact bearing designations only.

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TABLE V. Calibration of bore and outside diameter.

PN DES	BORE INCREMENTS	OD INCREMENTS
N	NO CALIBRATION	NO CALIBRATION
A	NO CALIBRATION	0.00010
B	NO CALIBRATION	0.00005
C	0.00010	NO CALIBRATION
D	0.00010	0.00010
E	0.00010	0.00005
F	0.00005	NO CALIBRATION
G	0.00005	0.00010
H	0.00005	0.00005

TABLE VI. Performance test.

PN DES	TYPE
1	MANUFACTURER'S STANDARD
2	STARTING TORQUE 1/

1/ Starting torque limits from table III of MIL-B-81793.

TABLE VII. Lubricant.

PN DES	SPECIFICATION
P	PRESERVATIVE 1/
A	MIL-L-6085
B	MIL-L-6085 2/
C	MIL-L-81846
D	MIL-L-81846 2/
E	MIL-G-23827
F	MIL-G-81322
G	MIL-G-81937
H	MIL-G-83261
J	MIL-S-81087
K	MIL-S-81087 2/

1/ PN DES "P" shall be used only with PN DES "P" in TABLE VIII.
2/ With barrier coat.

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TABLE VIII. Lubricant amount.

PN DES	QUANTITY
P	PRESERVATIVE AMOUNT AS REQUIRED BY MIL-P-197
1	MANUFACTURER'S STANDARD 1/
2	OIL PER TABLE II of MIL-B-81793
3	15% GREASE PER TABLE II of MIL-B-81793
4	25% GREASE PER TABLE II of MIL-B-81793
5	35% GREASE PER TABLE II of MIL-B-81793
6	45% GREASE PER TABLE II of MIL-B-81793

1/ Grease, fill to minimum 25%, maximum 40% of bearing void. Oil, immerse and then allow excess to drip off. The standard quantity of oil varies with each bearing size.

Custodians:
Army - AT
Navy - AS
Air Force - 99

Review activities:
Navy - SH
DLA - IS

User activities:
Navy - MC
Air Force - 84

Preparing activity:
Navy - AS

Agent:
DLS - IS

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